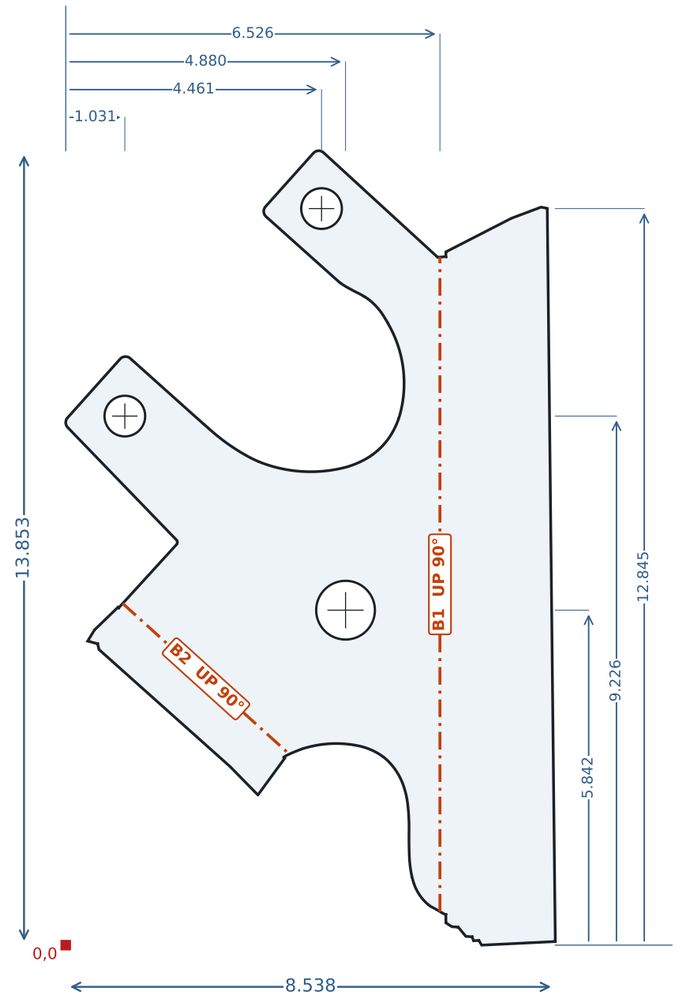
FR-1 · Plate with ears flat pattern (2 bends 90°)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 8.538″ × 13.853″ · ≈ 2.04 lb each · QTY 1 FR + 1 FL = 2 per vehicle

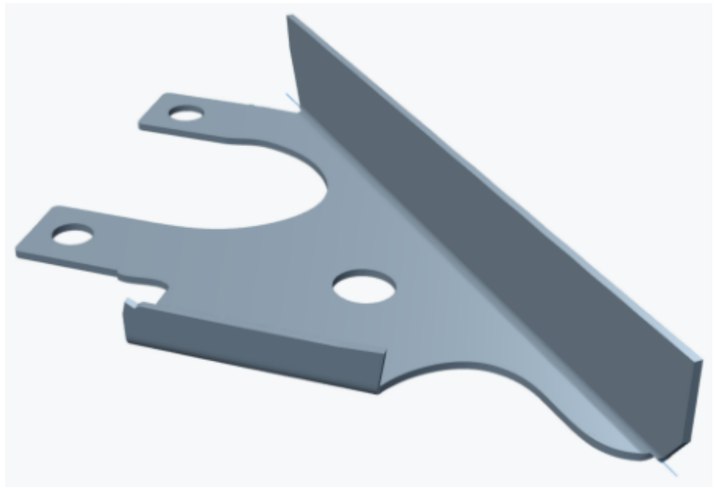
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 6.526
B2	90.0°	UP (toward)	R=t 0.120	2	(0.993, 5.959) (3.900, 3.331)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.709	1.031	9.226	laser
H2	Ø0.709	4.461	12.845	laser
H3	Ø1.024	4.880	5.842	laser

FILES

- FR-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-1_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* bend 90° up on dashed lines

FR-1 Side plate with ears and 2 flanges	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-1 · Plate with ears flat pattern (2 bends 90°)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**